

Citi Bike

Big Data Analytics Pipeline

Introduction to Big Data — Innopolis University

Team 19: Alex Kachmazov · Vladimir Rublev · Egor Sergeev · Anton Korotkov

May 2026

Agenda

01

Project Goal

What we set out to predict and why it matters

02

Dataset

2023 Citi Bike trip data — 10.2 M rows, 1.6 GB

03

Pipeline

4-stage end-to-end big data architecture

04

EDA Results

6 business insights from 10 M+ trips

05

ML Modeling

RF · SVM · Naive Bayes with grid search & CV

06

Dashboard

Apache Superset presentation & demo

07

Challenges

YARN, metastore, NB compatibility & more

Project Goal

Search — Exploratory Goal

Understand the spatial, temporal, and behavioural patterns of Citi Bike usage so that operations teams can allocate resources and marketing teams can target the right riders at the right moment.

6 EDA insights covering time, space & behaviour

ML — Predictive Goal

Given a trip's observable attributes — time, location, duration, bike type — predict whether the rider is a registered member or a casual user.

Why it matters:

- Personalised pricing for each segment
- Targeted member retention campaigns
- Tourist-area station planning

Dataset Characteristics

10.2 M

Total trips

1.6 GB

Raw CSV size

13

Raw features

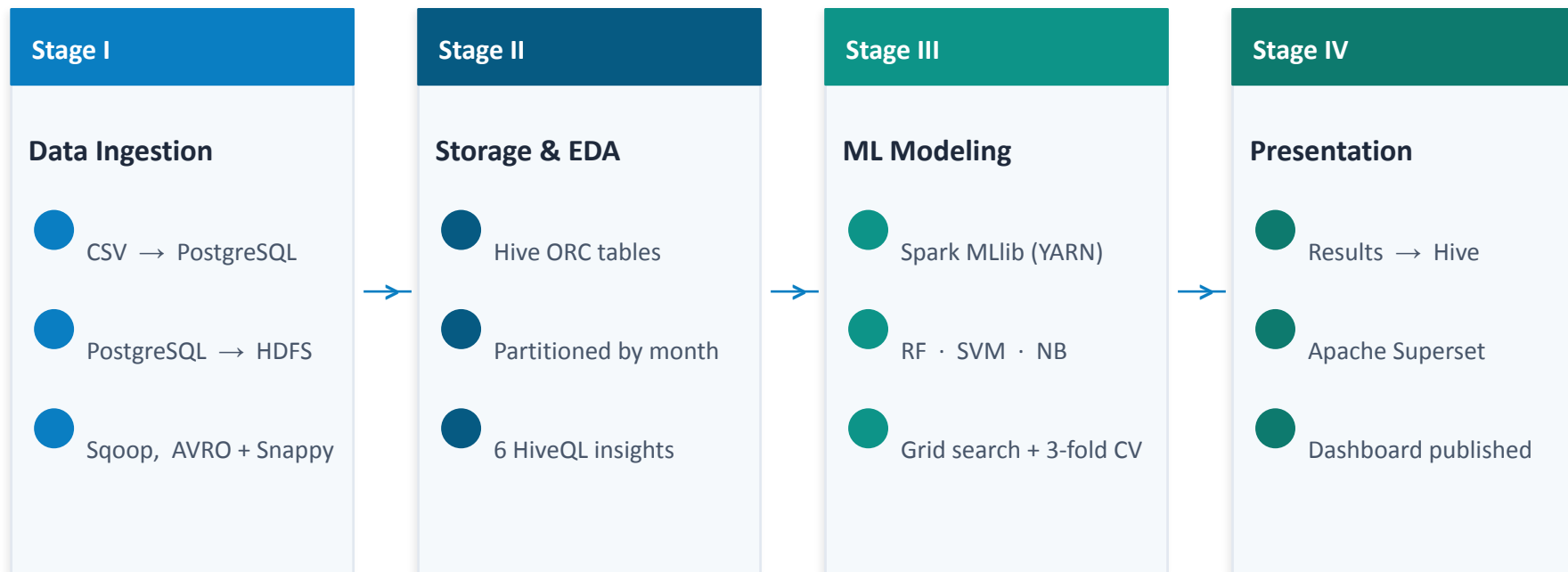
12 mo.

Jan – Dec 2023

Key Features

Column	Type	Description
ride_id	VARCHAR(50)	Unique trip identifier (Primary Key)
rideable_type	VARCHAR(50)	classic_bike / electric_bike / docked_bike
started_at / ended_at	TIMESTAMP	Trip start & end datetime (datetime feature)
start_lat/lng, end_lat/lng	DOUBLE PRECISION	GPS coordinates — departure & arrival (geospatial)
start_station_id / name	VARCHAR / TEXT	Station identifier and name
member_casual	VARCHAR(20)	Target label: member (1) or casual (0)

4-Stage Pipeline Architecture



All stages scripted & reproducible — run from repo root with: `bash scripts/stage*.sh`

Stage I & II: Ingestion → Hive

Stage I — Data Ingestion

- 1. Download** Monthly CSVs from Kaggle (1.6 GB)
- 2. PostgreSQL** citibike_trips table via COPY command
- 3. Sqoop → HDFS** Format: AVRO | Codec: Snappy
- 4. Outputs** .avsc + .java schema files in repo

Stage II — Hive Storage & EDA

- Table design:**
- Format: ORC + Snappy
 - Partition: ride_year, ride_month
 - Bucket: start_station_id (8 buckets)
- Cleaning:**
- Null rideable_type → 'unknown'
 - AVRO unix timestamps → datetime
 - Derived: ride_duration_min, ride_hour
 - Dropped invalid trips (end < start)

How We Analyzed the Data

Stage II — Exploratory Data Analysis with HiveQL on Tez engine

01 Hive Tables

Raw AVRO → ORC table with partitioning by month and bucketing by station for fast analytical scans.

02 6 HiveQL Queries

Each query targets a specific business question: time patterns, spatial demand, user mix, fleet.

03 CSV Export

Results saved as single-partition CSVs (q1.csv – q6.csv) from HDFS to the repository output/folder.

04 Visualisation

Python (matplotlib) charts generated from each CSV and saved as PNG images for report and Superset.

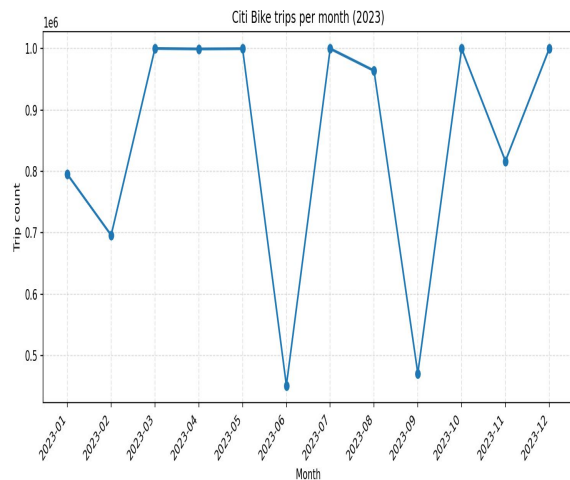
05 Data Stories

Each insight paired with a 3–5 sentence stakeholder narrative explaining the business implication.

06 Superset

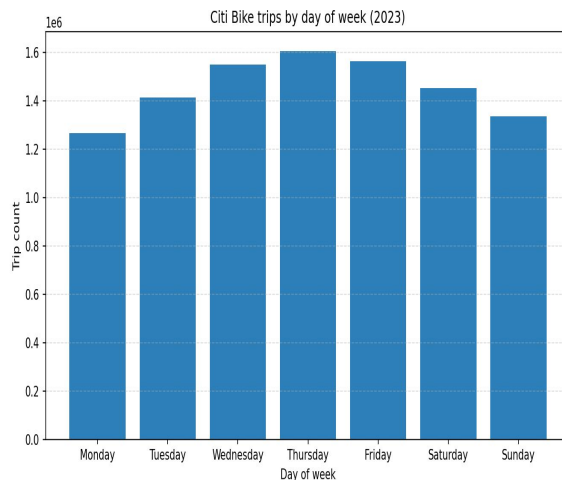
Pre-aggregated result tables loaded as Hive views, then charted in Apache Superset Tab 2.

EDA Results — Time Patterns



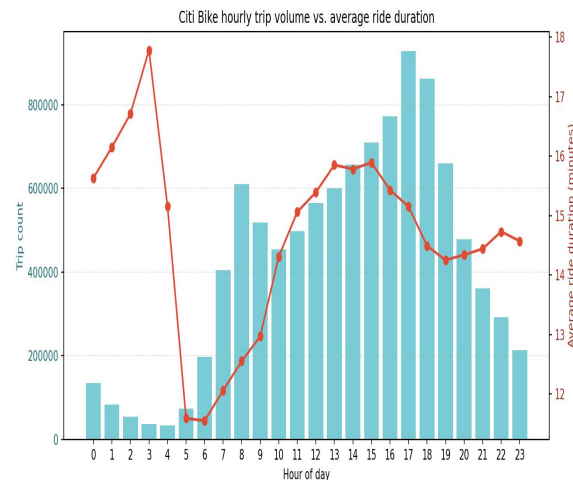
Monthly Trip Volume

Cold-weather dip in Jan–Feb. June & September show gaps — data quality issue in source archives.



Trips by Day of Week

Weekdays dominate volume (commuters). Weekends see fewer but longer trips — leisure riding.

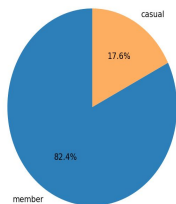


Hourly Volume vs Duration

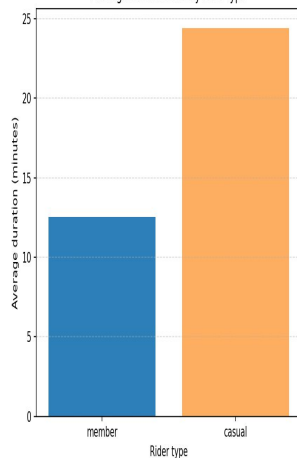
Sharp 8 AM and 6 PM peaks. Midday trips fewer but longer — guides rebalancing schedule.

EDA Results — Users, Stations & Fleet

Share of trips: member vs casual



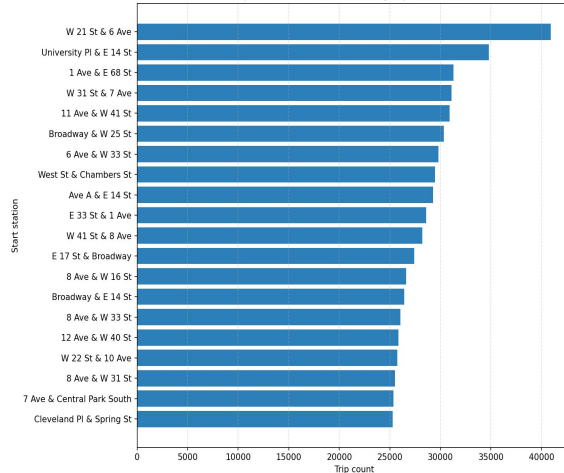
Average ride duration by rider type



Member vs Casual Mix

Members drive volume; casuals take longer trips.
Different revenue and retention dynamics.

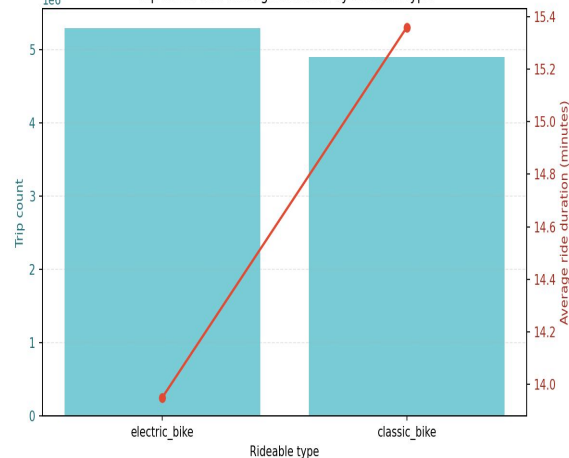
Top 20 Citi Bike start stations by trip count (2023)



Top 20 Start Stations

Demand highly concentrated in midtown & lower Manhattan hubs — priority for rebalancing.

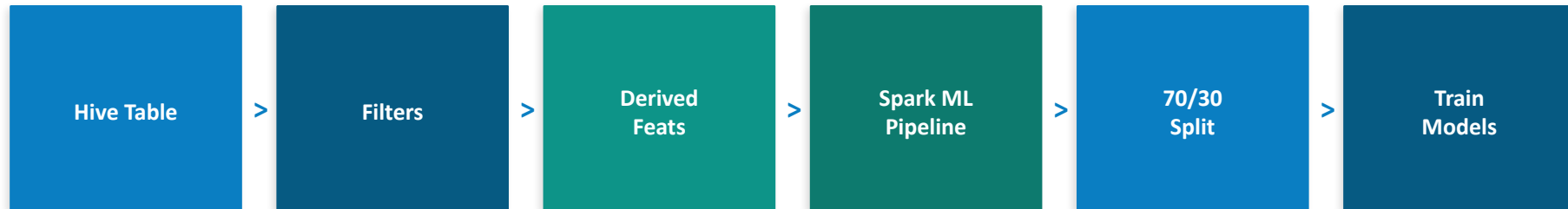
Trip count and average duration by rideable type



Rideable Type Usage

Classic bikes dominate. E-bikes shorter trips. Fleet investment should weigh demand vs. cost.

Stage III: Feature Engineering & Pipeline

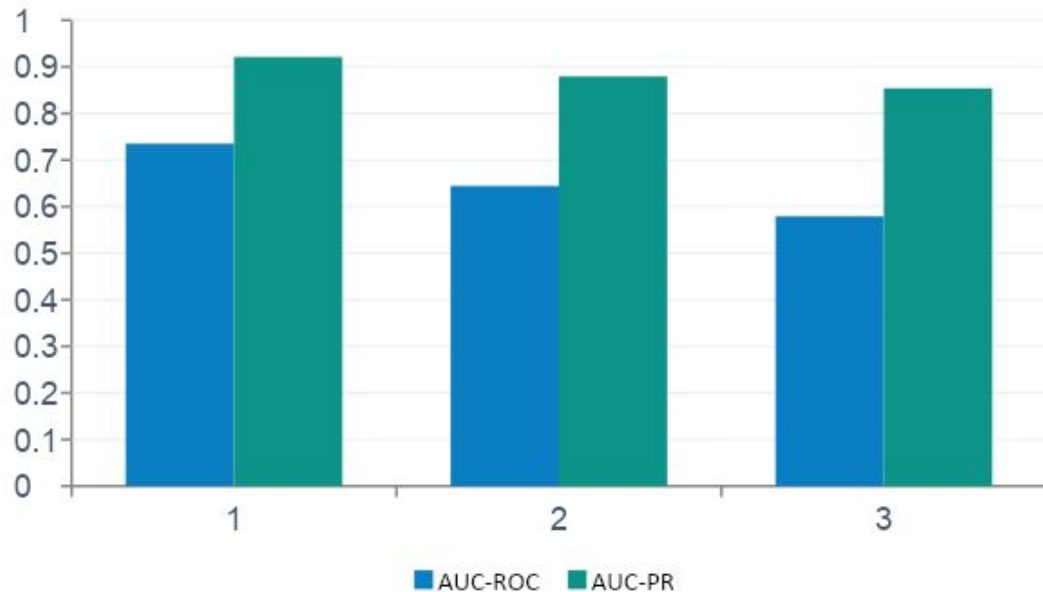


Features fed into the pipeline:

Numeric (continuous):	ride_duration_minutes, trip_distance_km (Haversine), start/end lat & lng
Numeric (cyclic):	ride_hour, ride_weekday, ride_month
Categorical → OHE:	rideable_type (3 values: classic, electric, docked)
Categorical → Index:	start_station_id (high-cardinality)
Binary:	is_weekend (derived: 1 if ride_weekday ∈ {6,7})
Target label:	member_casual → 1 (member) / 0 (casual)

Stage III: Model Evaluation Results

AUC-ROC & AUC-PR by Model



Random Forest ★ BEST

ROC: 0.7345 | PR: 0.9210 | ~39 min

Best: numTrees=100, maxDepth=15, minInstances=10

Linear SVM

ROC: 0.6440 | PR: 0.8795 | ~9 min

Best: regParam=0.001, tol=0.0001

Naive Bayes

ROC: 0.5793 | PR: 0.8538 | ~2 min

Best: smoothing=2.0, scaler_max=0.5

All models: 27 hyperparameter combos · 3-fold cross-validation · trained on YARN cluster

Best Model: Sample Prediction Demo

Random Forest — single trip inference

Input Features

<i>rideable_type</i>	classic_bike
<i>ride_hour</i>	18 (6 PM)
<i>ride_weekday</i>	2 (Tuesday)
<i>ride_month</i>	6 (June)
<i>ride_duration_minutes</i>	12.5
<i>trip_distance_km</i>	1.85
<i>is_weekend</i>	0 (weekday)

Prediction

MEMBER

Confidence: 86.6%

[0.134, 0.866] — $P(\text{casual}) \rightarrow P(\text{member})$

A 12.5-min Tuesday evening commute on a classic bike is highly characteristic of a registered member.

Stage IV: Apache Superset Dashboard

Team 19 — Citi Bike Big Data Pipeline | Published on Superset cluster

Tab 1 — Data Description

- Record count per table
- Column names & data types
- 20-row data sample

Tab 2 — Data Insights

- All 6 EDA charts from Stage II
- Pre-aggregated Hive result tables
- Filters by rider type / bike type

Tab 3 — ML Results

- AUC-ROC & AUC-PR bar charts
- Model comparison table
- Prediction distributions per model

Key finding: Members dominate volume · Casuals drive duration · RF is best classifier (AUC-ROC 0.735) · Station demand is heavily concentrated

Challenges Encountered

Hive Metastore Connectivity

PySpark couldn't reach the remote metastore without hive-site.xml on both driver and executor classpaths — not covered in course material.

Timestamp Parsing in AVRO

Sqoop serialised timestamps as integers with inconsistent scales (seconds vs. ms). Fixed with conditional unix_timestamp detection in the HQL insert.

Naive Bayes Compatibility

NB requires non-negative features. ChiSqSelector was incompatible with continuous inputs. Switched Bernoulli→Gaussian NB and tuned MinMaxScaler.

Data Gaps in Source Archive

The monthly 2023 extracts are inconsistent—some months sit near the usual cap while others (e.g., June and September) look like coverage gaps, so seasonality conclusions need a data-quality caveat.

Conclusion

■ End-to-end big data pipeline over 10.2 M Citi Bike trips — fully reproducible

■ 6 EDA insights: seasonal dips, rush-hour peaks, member/casual split, station hotspots

■ Random Forest wins: AUC-ROC 0.735, AUC-PR 0.921 (best of 3 x 27 hyperparameter combos)

■ Apache Superset dashboard published with 3 tabs covering data, insights, and ML results

■ Key insight: ride hour, trip distance, and start station are the strongest predictors

Thank you for your attention!!
